

Daniel Casadevall Jauhiainen

✉ dani.casadevall@gmail.com ☎ +34 645 555 013 🌐 github.com/danicasjau 🔗 linkedin.com/in/danielcasadevall

→ **Pipeline Technical Director** student targeting **R&D** and **Pipeline** roles in the VFX and animation industry, with a strong dual background in software development and **technical art**. Focused on automating production pipelines, optimizing **workflows**, and eliminating technical bottlenecks to empower artists.

SELECTED PROJECTS

I. Project Minivita - Interactive AI Character Framework September 2025

Developed a real-time **multimodal human representation** system integrating an interactive **SMPL-X motion diffusion** framework with an embeded vector database for dynamic character memory and centralized **LLM** execution. Optimized and quantized to run efficiently within consumer GPU memory limits. Rendered via **PySide OpenGL** viewport utilizing a **USD-based** geometry pipeline, implementing an asynchronous **96-frame** chunk system allowing low-latency **autonomous** digital character behavior.

II. AutoMayTex - AI-Driven Texturing Plug-in for Autodesk Maya March 2026 - Ongoing

Developed an **automated** diffusion texturing pipeline using **Maya API** running on a consumer GPU. Capturing multi-angle tetrahedral renders, and processing them via a local **FastAPI** server running **SDXL ControlNet**. Orchestrated via a **PySide6** GUI, it isolates the heavy runtime in a dedicated Python venv.

III. USD Pipeline Lead - ENCORE shortfilm Production - Prism September 2025 - Ongoing

Developed custom **Python** integration to embed **Pixar USD** asset variants and references seamlessly into the **Prism Pipeline** architecture. Deployed as the core pipeline for ENCORE (shortfilm, in production), connecting Houdini and Maya using Prism base code across all departments for a team of ~16 students.

IV. Procedural Generation Engine - Houdini Digital Assets (HDAs) 2025

Designed and implemented custom Houdini Digital Assets (HDAs) for **procedural** generation of urban architecture and complex **plant geometry**. Focused on building scalable team-node networks and exposed parameters tailored for layout artist integration.

V. tLoud Studio Manager - Production Management Web App February 2026 - Ongoing

Web-based asset task management platform for team production **workflows**. Asynchronous Python backend with a **Excel-driven** and **Discord** backend. Dynamic task assignments and multi-user progress monitoring. The project was made for organization of a **16 people** human team to work in ENCORE shortfilm project.

VI. Evolutive Biological Simulation - Neural Behavior Simulation 2022

Engineered a 2D biological simulation framework in Python to calculate **multi-agent** cell adaptation and behavioral inheritance across thousands of generations. Designed a **genetic pipeline** that compiled serialized **28-bit binary genomes** directly into modular sensory-input and motor-output execution loops. Coded entirely using the Python standard library.

EDUCATION

- 2023-2027 **Degree in Animation & VFX (In Progress)**
La Salle Bonanova University · Barcelona
- 2024 **Computer Science & Software Engineering (Paused)**
Universitat Oberta de Catalunya (UOC) · Barcelona
- 2022 **Technological Baccalaureate - Second Year**
La Sedeta Institute · Barcelona
- 2021 **High School (Lukio) - First Year**
Etelä-Tapiolan lukio · Helsinki

LANGUAGES

Catalan	Native · C2
Spanish	Native · C2
English	Upper · B2
Finnish	Native · C1

LANGUAGES & TECHNOLOGIES

- HTML / CSS JAVASCRIPT
- PYTHON SQL C++ C# C
- VEX PHP

PIPELINE & APIS

- PIXAR USD MAYA API
- PYSIDE6 / PYQT FASTAPI
- PRISM PIPELINE PYTORCH
- TRANSFORMERS DIFFUSERS

SOFTWARE

- AUTODESK MAYA SIDEFX HOUDINI
- BLENDER NUKE UNITY GIT
- LINUX

SOFT SKILLS

Ability to work in teams and adapt rapidly to dynamic environments. Strong communication skills with capacity for autonomous learning. Problem-solving oriented with independent judgment and a strong commitment to results.

ADDITIONAL EXPERIENCE

Youth Leader (Monitor de Lleure)
Agrupament Lluïsos de Gràcia (MEG)
Managed group dynamics, conflict resolution, and teamwork activities for youth education. Officially certified as a group coordinator.

Event Logistics & Visual Identity
2023 - 2025 | Cucafarra de Gràcia
Event organization, resource logistics, public relations, graphic layouts and cohesive visual identities for neighborhood cultural events.